

Autonomous Data Intelligence Report

Dataset Summary

Rows: 891

Columns: 12

Missing Values: 0

Duplicate Rows: 0

Problem Detection

Target Column: Survived

Problem Type: classification

Model Performance

Logistic Regression: 0.8212

Random Forest: 0.8212

XGBoost: 0.8268

Best Model: XGBoost

Analytics Summary

Here's an analysis: 1. **Dataset quality:** Excellent quality with 0 missing values and duplicates, indicating a clean dataset. The 891 rows provide a moderate size for robust modeling. 2. **Important patterns:** XGBoost's success points to complex, non-linear patterns. Survival likely involves interactions between features like gender, passenger class, and age, reflecting historical prioritization. 3. **Risks:** Moderate dataset size might limit generalization to broader populations. Inherent historical biases (e.g., social status, gender roles) within the data could lead to biased model outcomes if not interpreted carefully. 4. **Recommendation:** Leverage XGBoost's feature importance to uncover key survival drivers. Further, employ interpretability techniques (e.g., SHAP) to explain model decisions and derive deeper, actionable insights from these complex patterns.

AI Dataset Insights

1. Dataset Summary The Titanic dataset contains **891 rows** and **12 columns** (7 numeric, 5 categorical) targeting the binary classification of `Survived`. **Data Quality:** The dataset is fully preprocessed and clean, reporting **0% missing values** and **0 duplicate rows**, making it highly reliable for direct modeling. **Key Features:** Features span socio-economic status (`Pclass`, `Fare`), demographics (`Sex`, `Age`), family structures (`SibSp`, `Parch`), and metadata (`Name`, `Ticket`, `Cabin`, `Embarked`). ### 2. Important Patterns **Socio-Demographic Split:** Predictive signals in this dataset are heavily driven by the "women and children first" protocol (captured by `Sex` and `Age`) and socio-economic status (captured by `Pclass` and `Fare`). **Family Size Impact:** Combining sibling/spouse (`SibSp`) and parent/child (`Parch`) counts yields family-size metrics, which historically correlate with localized survival rates. **High-Cardinality Categoricals:** Features like `Name`, `Ticket`, and `Cabin` contain localized group survival patterns (e.g., families or cabin blocks) but require careful encoding to avoid overfitting. ### 3. Model Evaluation Three classifiers were trained and yielded highly competitive results: **Logistic Regression:** 82.12%

accuracy * **Random Forest:** 82.12% accuracy * **XGBoost (Best Model):** 82.68% accuracy

Analysis: XGBoost marginally outperformed the other models, likely due to its ability to capture non-linear relationships and feature interactions (e.g., `Sex` combined with `Pclass`). The tight clustering of scores (~82.1% to 82.7%) indicates a robust feature set and stable training. ### 4.

Recommendations

- * **Deploy Best Model:** Implement the saved XGBoost artifact (`ml\_engine/models/XGBoost.pkl`) for production or evaluation pipelines.
- * **Advanced Feature Engineering:**
 - * Extract titles (e.g., Mr., Mrs., Master) from the `Name` column to capture age/social status interactions.
 - * Extract deck letters from the `Cabin` column to capture physical proximity to lifeboats.
- * **Hyperparameter Tuning:** Run a Bayesian optimization (Optuna) or grid search specifically on the XGBoost model (focusing on `max\_depth`, `learning\_rate`, and `subsample`) to push accuracy past the ~83% threshold.

Model Selection Reasoning

XGBoost won because it achieved the highest score (0.8268), making it slightly more accurate in predicting who 'Survived'. Logistic Regression and Random Forest performed slightly worse, both scoring 0.8212. This indicates they were marginally less effective on this dataset. XGBoost often leverages complex patterns more effectively.